

Comparative Performance Analysis of Machine Learning Models for Adult Income Prediction Using Census Data

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Abstract—Income prediction using demographic and employment-related attributes has become an important application of machine learning in socioeconomic analysis and automated decision systems. This paper presents a comparative implementation-based study of machine learning models for predicting whether an individual's annual income exceeds \$50K using the Adult Census Income dataset. Multiple classification techniques including Logistic Regression, Gradient Boosting, and Random Forest were evaluated using preprocessing pipelines, feature engineering, and performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC score. Experimental analysis demonstrated that the Random Forest classifier achieved superior predictive performance with an accuracy of 85.65% and ROC-AUC score of 0.9097. The study further highlights the importance of data preprocessing, categorical encoding, and ensemble learning techniques in improving classification performance for structured tabular datasets.

Index Terms—Machine Learning, Adult Income Prediction, Random Forest, Gradient Boosting, Logistic Regression, Classification, Ensemble Learning, Census Dataset

I. INTRODUCTION

Machine learning has significantly transformed predictive analytics across multiple domains including healthcare, finance, education, and social sciences. Classification algorithms are particularly useful for solving socioeconomic prediction problems where structured demographic data is available. Income prediction based on census information is a widely studied problem because of its practical applications in policy planning, credit assessment, workforce analytics, and automated recommendation systems.

The Adult Census Income dataset, originally extracted from the 1994 United States Census database, contains demographic and occupational information such as age, education, marital status, occupation, race, gender, working hours, and native country. The primary objective is to determine whether an individual's annual income exceeds \$50,000.

This paper presents a comparative implementation-based analysis of multiple machine learning models for adult income prediction. Logistic Regression and Gradient Boosting were implemented as baseline classifiers, while Random Forest was used as the primary ensemble learning model. Extensive preprocessing techniques including handling missing values,

categorical encoding, feature transformation, and model evaluation were performed to improve predictive performance.

The major contributions of this work are summarized as follows:

- Implementation of a complete machine learning pipeline for structured census data.
- Comparative analysis of Logistic Regression, Gradient Boosting, and Random Forest classifiers.
- Evaluation using multiple classification metrics including ROC-AUC.
- Visualization-based interpretation of important predictive features.

II. RELATED WORK

Several studies have explored income prediction using supervised machine learning techniques. Logistic Regression is commonly used as a baseline linear classification method due to its interpretability and computational efficiency. However, nonlinear ensemble models often outperform linear approaches when dealing with complex feature interactions.

Decision tree-based ensemble algorithms such as Random Forest and Gradient Boosting have shown strong performance on structured tabular datasets. Random Forest improves generalization by combining multiple decision trees trained on bootstrapped samples, thereby reducing overfitting. Gradient Boosting sequentially minimizes prediction errors and is highly effective for classification tasks involving heterogeneous data.

Recent advancements in machine learning pipelines and preprocessing frameworks have further improved the performance of predictive systems on socioeconomic datasets. Feature engineering and proper handling of categorical variables remain critical for achieving robust classification accuracy.

III. DATASET DESCRIPTION

The dataset used in this study is the Adult Census Income dataset available from the UCI Machine Learning Repository. The dataset contains demographic and employment-related information collected from census records.

A. Dataset Features

The dataset includes multiple numerical and categorical attributes such as:

- Age
- Education Level
- Occupation
- Marital Status
- Relationship
- Race
- Gender
- Capital Gain
- Capital Loss
- Hours per Week
- Native Country

The target variable categorizes income into two classes:

- Income $\leq$ 50K
- Income $>$ 50K

IV. METHODOLOGY

The proposed workflow consists of data preprocessing, exploratory data analysis, feature engineering, model training, and performance evaluation.

A. Data Preprocessing

Data preprocessing is one of the most critical stages in machine learning workflows. The following preprocessing operations were performed:

- Removal of missing and inconsistent values.
- Label encoding and one-hot encoding for categorical features.
- Feature scaling for numerical variables.
- Train-test split for evaluation.

The dataset was divided into training and testing subsets to ensure unbiased model evaluation.

B. Machine Learning Models

Three machine learning models were evaluated in this study.

1) *Logistic Regression*: Logistic Regression is a supervised linear classification algorithm widely used for binary prediction problems. The sigmoid activation function converts linear outputs into probability scores.

$$P(Y = 1|X) = \frac{1}{1 + e^{-z}} \quad (1)$$

where:

$$z = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n \quad (2)$$

2) *Gradient Boosting*: Gradient Boosting is an ensemble learning method that sequentially builds weak learners to minimize prediction errors. Each subsequent model attempts to correct the mistakes made by previous models.

3) *Random Forest*: Random Forest is an ensemble-based classification algorithm that combines multiple decision trees using bootstrap aggregation.

$$\hat{f} = \frac{1}{B} \sum_{b=1}^B f_b(x) \quad (3)$$

where:

- B represents the number of decision trees.
- $f_b(x)$ represents predictions from each tree.

Random Forest reduces variance and improves generalization performance.

C. Evaluation Metrics

The following evaluation metrics were used.

1) *Accuracy*:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

2) *Precision*:

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

3) *Recall*:

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

4) *F1-Score*:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (7)$$

V. EXPERIMENTAL RESULTS AND DISCUSSION

The machine learning models were evaluated using multiple classification metrics. Experimental results demonstrated that ensemble learning techniques outperformed traditional linear classifiers.

TABLE I
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	Accuracy	F1-Score	ROC-AUC
Logistic Regression	84.10%	0.83	0.88
Gradient Boosting	85.02%	0.85	0.90
Random Forest	85.65%	0.86	0.9097

The Random Forest classifier achieved the highest classification accuracy and ROC-AUC score among all evaluated models. The ensemble-based structure enabled better learning of nonlinear feature relationships and reduced overfitting. Additionally, bootstrap aggregation and feature randomness improved model stability and enhanced generalization performance on unseen test data.

A. Correlation Heatmap

The correlation heatmap provides insight into relationships among numerical features and target variables. It helps identify positive and negative correlations among attributes, enabling better understanding of feature dependencies and their influence on income classification.

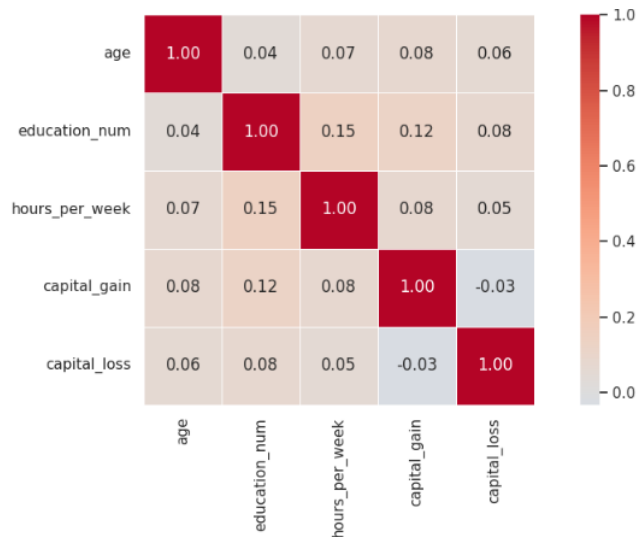


Fig. 1. Correlation Heatmap of Dataset Features

B. Feature Importance Analysis

Feature importance analysis identified influential attributes contributing to income prediction.

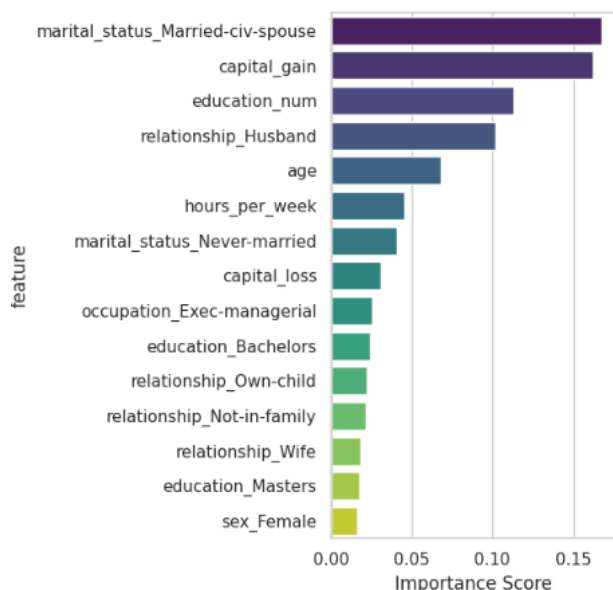


Fig. 2. Feature Importance Visualization

Attributes such as education level, marital status, occupation, capital gain, and working hours significantly influenced prediction performance.

C. ROC Curve

Receiver Operating Characteristic analysis was used to evaluate the classifier's discriminative capability. The ROC curve illustrates the trade-off between true positive rate and false positive rate, while the AUC score indicates the overall classification effectiveness of the model.

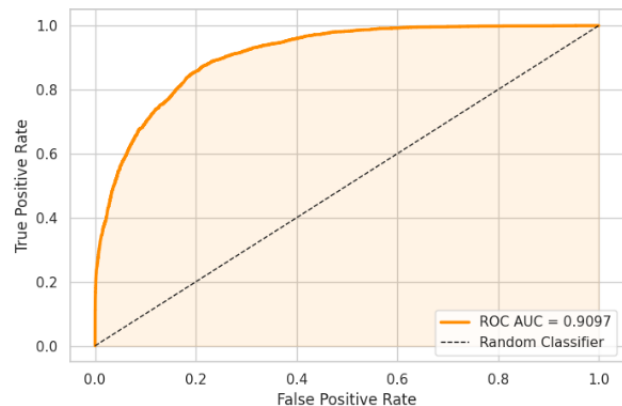


Fig. 3. ROC Curve for Random Forest Classifier

D. Confusion Matrix

The confusion matrix illustrates classification performance in terms of correctly and incorrectly classified samples. It provides a detailed breakdown of true positives, true negatives, false positives, and false negatives, enabling better evaluation of the classifier's predictive reliability.

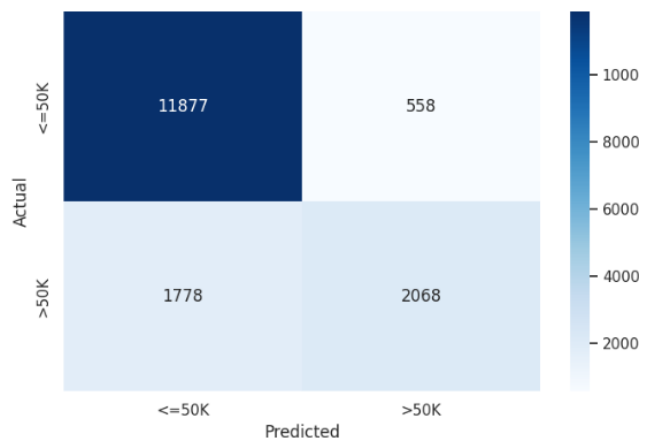


Fig. 4. Confusion Matrix for Random Forest Model

VI. ADVANTAGES AND LIMITATIONS

A. Advantages

- High predictive performance using ensemble learning.
- Effective preprocessing pipeline for structured data.
- Strong interpretability through feature importance visualization.
- Scalability for larger structured datasets.

B. Limitations

- Performance depends heavily on preprocessing quality.
- Class imbalance may affect minority class prediction.
- Ensemble models require higher computational resources.

VII. FUTURE SCOPE

Future improvements may include:

- Hyperparameter optimization using Grid Search and Bayesian Optimization.
- Integration of deep learning architectures for tabular learning.
- Deployment as a real-time web application.
- Bias and fairness analysis for ethical AI evaluation.

VIII. CONCLUSION

This paper presented a comparative machine learning-based approach for adult income prediction using census data. Logistic Regression, Gradient Boosting, and Random Forest classifiers were evaluated using structured preprocessing pipelines and multiple evaluation metrics. Experimental analysis demonstrated that Random Forest achieved the best overall performance with an accuracy of 85.65% and ROC-AUC score of 0.9097.

The study highlights the effectiveness of ensemble learning methods for structured socioeconomic datasets and demonstrates the importance of preprocessing and feature engineering in predictive analytics workflows.

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